

Note: Let $\langle G, * \rangle$ be an Abelian group and H be a subgroup of G . Then $\Rightarrow a^{-1}Ha = H$
 $aH = Ha$ for every $a \in G$ OR $\Rightarrow aHa^{-1} = H$
(that is left and right cosets of H are equal)

Defⁿ: A subgroup $\langle H, * \rangle$ of a group $\langle G, * \rangle$ is called a normal subgroup if
 $a^{-1}Ha \in H$ for every $a \in G$
(OR if $a^{-1}ha \in H$ for any $a \in G$ and $h \in H$)

Ex: Prove that every subgroup of an Abelian group is normal (NTU 2020 4M)

Solⁿ: $\langle G, * \rangle \rightarrow$ Abelian (Given)

$H \subseteq G$ $H \rightarrow$ Normal (To prove)

$\rightarrow$ we will prove $a^{-1}ha \in H$ for any $a \in G$ & for any $h \in H$.

$\rightarrow$ Let $a \in G, h \in H$

$$\begin{aligned} a^{-1}ha &= a^{-1}(ha) = a^{-1}(ah) \quad (\because G \text{ is abelian}) \\ &= (a^{-1}a)h \quad (\because \text{using associative prop}) \\ &= eh \\ &= h \in H \end{aligned}$$

$$\therefore \boxed{a^{-1}ha \in H}$$

$\therefore$ Every subgroup of an Abelian group is normal.

Note: Let $\langle G, * \rangle$ be a group. We know that G always has at least two subgroups $\langle \{e\}, * \rangle$ and $\langle G, * \rangle$ itself. These two subgroups are called trivial subgroups.

$\rightarrow$ Trivial subgroups are always normal.

Ex: Consider the group G of 2×2 matrices with rational entries and non zero determinant. Let H be the subset of G consisting of matrices whose upper-right entry is zero, i.e. matrices of the form $\begin{bmatrix} a & 0 \\ c & d \end{bmatrix}$ check whether H is a normal subgroup of G or not.

$$\text{sol}^n \Rightarrow G = \left\{ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} / a, b, c, d \in \mathbb{Q} \ \& \ |A| \neq 0 \right\}$$

$$H = \left\{ \begin{bmatrix} a & 0 \\ c & d \end{bmatrix} \in G \right\}$$

→ No, H is not a normal subgroup of G .
that means we prove $A^{-1}DA \notin H$ for any $A \in G$ and $D \in H$

$$\rightarrow \text{Let } A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \Rightarrow |A| = 3 - 2 = 1 \neq 0 \therefore A \in G$$

$$D = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \in H$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

$$\left(\begin{array}{l} * A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ \text{then } A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \end{array} \right)$$

$$\rightarrow \text{Now } A^{-1}DA = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1+0 & 2+0 \\ 1+1 & 2+3 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 3-4 & 6-10 \\ -1+2 & -2+5 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -4 \\ 1 & 3 \end{bmatrix} \notin H$$

$\therefore H$ is not a normal subgroup of G .

Ex: Let $G = \langle \mathbb{R}, + \rangle$ and $H = \langle \mathbb{R}^+, \times \rangle$. Define $f: G \rightarrow H$ by $f(x) = 2^x$; prove that f is a homomorphism.

Solⁿ For any $a, b \in G$

$$f(a+b) = 2^{a+b} = 2^a \times 2^b = f(a) \times f(b)$$

$\therefore f: G \rightarrow H$ is a homomorphism.

Ex: Let G be the group of non-zero complex numbers under multiplication and let G' be the group of non-zero real numbers under multiplication. Prove that the mapping $f: G \rightarrow G'$ by $f(z) = |z|$ is a homomorphism.

Solⁿ $G = \mathbb{C} - \{0\}$, $G' = \mathbb{R} - \{0\}$

$\rightarrow$ To prove: For any $z \in G$, $f(z) = |z|$ is a homomorphism (i.e. $f(z.w) = f(z) \cdot f(w)$)

$\rightarrow$ Let $z, w \in \mathbb{C} - \{0\} = G$

$$f(z.w) = |z.w| = |z| \cdot |w| = f(z) \cdot f(w) \text{ for any } z, w \in G$$

$\therefore f(z) = |z|$ is a homomorphism.

Ex: Let a be any element in a group G . Then function $f: \mathbb{Z} \rightarrow G$ defined by $f(n) = a^n$ is a homomorphism. $\langle \mathbb{Z}, + \rangle$ and $\langle G, * \rangle$

Solⁿ Given $a \in G$

Let $m, n \in \mathbb{Z}$

$$f(m+n) = a^{m+n} = a^m * a^n = f(m) * f(n)$$

$\therefore f$ is a homomorphism

Ex: Let $\langle G, * \rangle$ and $\langle H, \Delta \rangle$ be two groups with identities e_G and e_H respectively. Suppose $f: G \rightarrow H$ be a group homomorphism. Prove that (a) $f(e_G) = e_H$ (b) $f(a^{-1}) = f(a)^{-1}; a \in G$

Solⁿ: (a) Since e_G is the identity of G ,

$$e_G = e_G * e_G$$

$$\therefore f(e_G) = f(e_G * e_G)$$

$$f(e_G) = f(e_G) \Delta f(e_G) \quad (\because f: G \rightarrow H \text{ is a homomorphism})$$

$$\underline{f(e_G) \Delta f(e_G)^{-1} = f(e_G) \Delta f(e_G) \Delta f(e_G)^{-1}}$$

$$\therefore e_H = f(e_G) \Delta e_H$$

$$\boxed{e_H = f(e_G)} \quad (\because e_H \text{ is the identity})$$

(b) Let $a \in G$ ~~then~~ $a * a^{-1} = a^{-1} * a = e_G$

$$\therefore f(a * a^{-1}) = f(a^{-1} * a) = f(e_G)$$
 Since f is a homomorphism

$$\therefore f(a) \Delta f(a^{-1}) = f(a^{-1}) \Delta f(a) = e_H$$

$$\therefore f(a)^{-1} \text{ is the inverse of } f(a)$$

$$\therefore \boxed{f(a)^{-1} = f(a^{-1})}$$

Ex: Show that $\langle G, +_5 \rangle$ is a cyclic group,
where $G = \{0, 1, 2, 3, 4\}$ (UTU 2020)

Solⁿ: $G = \{0, 1, 2, 3, 4\} = \{[0], [1], [2], [3], [4]\}$

$[a] +_5 [b] = [\text{Remainder when } a+b \text{ is divided by } 5]$

$$[1]^1 = [1] \quad \checkmark$$

$$[1]^2 = [1] +_5 [1] = [(1+1) \pmod{5}] = [2 \pmod{5}] = [2]$$

$\therefore [2] = [1]^2 \quad \checkmark$

$$[1]^3 = [1]^2 +_5 [1] = [2] +_5 [1] = [3 \pmod{5}] = [3]$$

$\therefore [3] = [1]^3 \quad \checkmark$

$$[1]^4 = [1]^3 +_5 [1] = [3] +_5 [1] = [4 \pmod{5}] = [4]$$

$\therefore [4] = [1]^4 \quad \checkmark$

$$[1]^5 = [1]^4 +_5 [1] = [4] +_5 [1] = [5 \pmod{5}] = [0]$$

$\therefore [0] = [1]^5 \quad \checkmark$

$\therefore [1]$ is the generator of G .

Every element is represented in power of 1.

$\therefore G$ is a cyclic group.

Note: For any $m \in \mathbb{N}$, $(\mathbb{Z}_m, +_m)$ is a cyclic group with generator $[1]$.